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RESEARCH ARTICLE

Developing a statistical approach of evaluating daily maximum and minimum temperature observations from third-party automatic weather stations in Australia

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Abstract

Third-party automatic weather stations (TPAWS) provide a compelling data source for scientists and practitioners to observe and estimate more accurate fine-scale atmospheric conditions, including daily maximum and minimum temperature (denoted as T_{max} and T_{min}, respectively), than the current primary weather observation network can offer. Several uncertainties and errors arise in data from TPAWS as the quality control applied to these stations may be inadequate or ad hoc. In this study, we develop a statistical approach to evaluate the quality of daily T_{max} and T_{min} observations collected from TPAWS in Australia. Our approach compares a target observation with multiple types of reliable reference data, including neighbouring primary weather observations from the official Bureau of Meteorology of Australia stations, Australian Gridded Climate Data, and numerical weather prediction data. Guided by the operational requirements in terms of automation, interpretability, and simplicity as well as expandability, a separate test is formed for each type of reference data and then all the individual tests are merged to generate a single result based on a Gaussian mixture model that is used to provide the final overall assessment for each TPAWS observation. The overall assessment is made in the form of a *p*-value-based confidence score that measures the difference between the target observation and trusted reference data. Our method is validated by synthetic datasets based on high-quality observations and is also applied to daily T_{max} and T_{min} observations from 184 TPAWS owned by the Department of Primary Industries and Regional Development of Western Australia. The framework can be readily applied to different regions with different reliable or trusted data sources.

KEYWORDS

daily maximum temperature, daily minimum temperature, data quality, statistical approach, third-party automatic weather stations

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1 | INTRODUCTION

Accurate weather observations, also commonly referred to as measurements (we do not differentiate between the two in this study), are essential for developing and testing climate and weather models, which form the basis of forecasting and prediction over time (temporal forecasting in the future) and space (spatial forecasting at unknown locations). These observations also provide inputs to downstream applications such as hydrological models and agricultural models, which help end users with decision making. Relevant tasks include irrigation and herbicide application planning, management of weather changes and natural hazards, and preparation for long-term challenges such as climate change. The weather observations or their estimated values from trusted sources are also the foundations for index-based or parametric insurance products for agriculture to ensure food security (see, for e.g., Greatrex et al., 2015).

Traditionally, weather observations are obtained from the weather stations managed by government agencies, which are referred to as primary weather stations. A suite of quality control (QC) and quality assurance (QA) procedures are implemented by experts to ensure that these observations are of the highest possible quality. However, primary weather stations can be sparsely distributed, resulting in limited local details. For example, there are currently less than 700 automatic weather stations (AWS) operated by the Bureau of Meteorology (BoM) of Australia on the Australian mainland and Tasmania to provide real-time temperature observations as of 2020. For temperature, which is the focus of this paper, the observations collected by BoM AWS do not have sufficient spatial coverage for a good representation of weather events in remote areas such as farmlands. End users thus tend to rely on heavily interpolated grid data products such as the Australian Gridded Climate Data (AGCD) with a grid size of 5 km × 5 km (Jones et al., 2009).

In order to satisfy their own requirements, and by taking advantage of technology advancements with affordable AWS, more and more end users have established their own observation networks of AWS (which are called third-party weather networks in this study) to monitor weather in a cost-effective way (Buytaert et al., 2014; Muller et al., 2015). In Australia, third-party weather networks have approx. 8000 third-party AWS (TPAWS) and can provide temperature observations at many more locations than the BoM networks. Some third-party weather networks (e.g., Netatmo and Weather Underground) use a distributed network of individuals to obtain the data from public sensors connected by the internet, which are referred to as crowdsourced weather networks (also known as private weather network [Chen, Saunders, & Whan, 2021] or citizen weather network ([Napoly et al., 2018] in the literature). Other third-party weather networks are managed and operated by local government agencies, organisations or reliable private network providers, which are referred to as corporate weather networks. As an example of Australian third-party networks, the Department of Primary Industries and Regional Development (DPIRD) of Western Australia (WA) provides a network of approx. 200 TPAWS throughout the state to provide timely, relevant and local weather data to assist growers and regional communities for better decision making (DPIRD, 2022a).

The growing TPAWS could potentially offer a solution to extend the coverage of the existing primary networks while supplying weather information for their own usages. The inclusion of TPAWS can significantly increase the density of the weather observation network. For example, in southwest Western Australia, the average number of AWS within a radius of 20 km increases nearly threefold from 0.36 to 1.12 with the inclusion of the DPIRD stations. Figure S1 further shows the spatial distribution of the network density.

However, the most critical challenge with TPAWS is ensuring that weather data collected have sufficient quality, especially for crowdsourced weather networks (Bell et al., 2013, 2015; de Vos et al., 2017, 2019; Meier et al., 2017). Multiple factors contribute to the inaccuracy observed in temperature readings from TPAWS. (1) In instances of inadequate ventilation during sunny, low-wind conditions, the economical temperature sensors utilised by TPAWS might be excessively exposed to solar irradiation, leading to heat-biased readings beyond the expected range (Lee et al., 2018a, 2018b; Sun et al., 2015; Wang et al., 2018). Some prior research has aimed to rectify the influence of solar radiation on temperature biases. (2) The majority of TPAWS installations have not adhered to the international standards set forth by the World Meteorological Organization (World Meteorological Organization, 2018). The incorrect positioning of temperature sensors is a common issue among TPAWS in crowdsourced weather networks. For instance, placing stations adjacent to building walls can consistently yield inaccurate readings. (3) Low-cost devices employed in TPAWS may not undergo regular maintenance routines (such as periodic calibration to rectify sensor drift over time), rendering them more susceptible to generating inaccurate readings, instrumental errors, and processing errors compared to primary weather stations. (4) Malicious entities might intentionally manipulate data or sensors to produce misleading information. There have been reported instances of adversaries tampering with TPAWS to support fraudulent insurance claims (source: https://indianexpress.com/article/cities/pune/rain-gauge-fraudulent-insurance

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-claims-pradhan-mantri-fasal-bima-yojana-8986923/).
In practice, a QA step is essential to filter out obviously low-quality TPAWS. The work presented in our paper focuses on TPAWS that maintain reasonable levels of data quality (i.e., partially trusted weather stations).

When end users may use their own third-party weather data for decision making, for example, in the case of index-based insurance, the data may be used as ground truth. For these reasons, ensuring correct, timely, and trusted assessment of these third-party weather data is critical. Proper QC and QA methods should thus be developed to evaluate the confidence level of an observation, which can be potentially used to identify and remove erroneous data according to some criteria. While QA is a one-off exercise to screen untrusted stations by experienced personnel, more research studies have focused on the development of QC methods. A wealth of research into QC methods for traditional weather observations is available in the literature (Bertrand et al., 2013; Feng et al., 2004; Fiebrich et al., 2010; Gandin, 1988; Rusticucci & Renom, 2008; Stepanek et al., 2009; You & Hubbard, 2006). More recently, some researchers have developed QC methods specifically for the temperature data from third-party weather networks. Meier et al. (2017) designed a sequence of QC levels to assess crowdsourced weather observations for urban climate research. Napoly et al. (2018) proposed a statistically based QC routine to filter out erroneous temperature observations from crowdsourced weather data and applied it to Netatmo private weather stations in two European cities. This routine has seven QC levels and can efficiently detect doubtful data due to solar exposition and sitting issues. Fenner et al. (2021) further extended the QC method developed by Napoly et al. (2018) with two additional QC levels. Chakraborty et al. (2020) introduced a statistical metric to assess the quality of crowdsourced weather observations using a coarser but high-quality reference data. Other recent studies into the QC of crowdsourced weather observations are available for application to rainfall (Bardossy et al., 2021; Chen, Behl, & Goodall, 2021; de Vos et al., 2019) and wind (Chen, Saunders, & Whan, 2021; Droste et al., 2020).

All these QC methods developed for TPAWS (mostly crowdsourced) largely depend on a comparison with the observations from neighbouring primary weather stations. However, the fact is that many TPAWS are typically intended to fill the gap of the primary networks in many cases where such primary weather observations are not available. This makes these QC methods effective in only relatively small-scale regions, such as urban climate research (Meier et al., 2017). Therefore, there is a need and opportunity for developing new methods to evaluate the data quality of TPAWS in a large-scale region where there are limited or no neighbouring primary weather stations.

In this manuscript, we develop a data-driven framework to assess the quality of daily maximum and minimum temperature (T_{max} and T_{min}) observations collected from TPAWS in Australia by considering multiple types of trusted reference data, including neighbouring BoM weather observations, gridded climate data, and numerical weather prediction (NWP) data. Although temperature is often observed at higher frequencies from minutes to hours, daily T_{max} and T_{min} are typically reported in different datasets and are primarily important for end users.

In the development of our assessment procedure, the following key requirements are considered. (1) Automation: the fully AWS in third-party networks generate a large number of observations fed into a system at real time or near real time, making it impossible to conduct QC by any experts manually. Therefore, an assessment procedure needs to be fully automatic or with minimal and feasible human intervention. (2) Interpretability: if the potential procedure mainly consists of data-driven methods, then the procedure's interpretability is crucial for auditing purposes. This is considered essential by both the scientific community and industry (see, for e.g., Horkoff, 2019 and Zhu et al., 2022). (3) Expandability and simplicity: from software and operational point of view, the developed framework should be easily expanded to test and then incorporate new sources of trusted data once they are available.

Given these requirements, our proposed procedure/framework is designed to evaluate the data quality from different perspectives and can still be applicable in the situation where a certain type of reference data (e.g., neighbouring primary weather observations) are not readily available. Each type of reference data forms an individual test in two stages: (1) issue predictions of daily T_{max} and T_{min} for every target TPAWS observation and (2) quantify the associated prediction uncertainty. The primary assumption of our evaluation is that TPAWS observations are mostly accurate and have errors in the form of outliers. An overall assessment is made by combing the test results from different individual tests and assigns a confidence score statistically to each TPAWS observation based on its agreement with available reference data. The use of confidence scores, rather than traditional quality flags, enables end users to make their own assessment of risk and fit for purpose.

The remainder of this manuscript is organised as follows. Section 2 introduces the reference data used in the test methods. Section 3 describes the details of each individual test and the method to combine them together. Section 4 validates the test method based on synthetic datasets and Section 5 provides a case study based on 184 TPAWS from the DPIRD network. Conclusions are made in Sections 6.

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2 | REFERENCE DATA

The following three types of reference data are used in this study to evaluate the quality of daily T_{max} and T_{min} observations collected from TPAWS in Australia:

1. BoM observations. BoM observations refer to high-quality daily T_{max} and T_{min} observations from the BoM networks (primary networks) and provide the most accurate measurements of daily T_{max} and T_{min} (defined by the maximum temperature in the 24 hours from 0900 AM and the minimum temperature in the 24 hours to 0900 AM, respectively). The BoM networks consist of approx. 700 AWS, which generate and handle data automatically. Some quality checking has been undertaken and quality flags have been appended to the BoM observations. Any observations with the quality flags of 'suspect', 'inconsistent' or 'wrong' are excluded from the reference data. The BoM manual weather stations where the daily T_{max} and T_{min} are manually recorded by an observer reading a liquid-in-glass thermometer are not included in this study because most of them cannot provide data in a 'real-time' manner. To summarise, our study solely relies on weather observations from AWS, excluding manual observations. The BoM weather observations can be requested from the BoM Climate Data Online (Bureau of Meteorology, 2022c).
2. ACCESS NWP forecasts. The Australian Community Climate and Earth-System Simulator (ACCESS) NWP (Puri et al., 2013) data are made by operational NWP systems employed by the BoM. The ACCESS NWP systems adopt the Unified Model/Variational Assimilation (UM/VAR) system developed by the United Kingdom Met Office (UKMO). This study uses the ACCESS system APS2 (Australian Parallel Suite version 2; see Bureau of Meteorology, 2017) for reference, which covers the study period in the synthetic data example (Section 4) and the case study based on the DPIRD network (Section 5). T_{max} and T_{min} forecasts at hourly intervals up to 240 hours are issued by the ACCESS systems four times daily (0000, 0600, 1200 and 1800 UTC) with approx. 12 km × 12 km resolution. The ACCESS NWP APS2 forecast data are available from 07 June 2016 to 25 September 2020 and can be directly downloaded from National Computational Infrastructure (NCI) with an OpenDAP server option (NCI, 2022a). More information about the ACCESS NWP data can be found at Bureau of Meteorology (2022a).
3. AGCD, The Australian Gridded Climate Data (AGCD) is the BoM's official dataset for climate analyses covering the weather variables of rainfall, maximum temperature, minimum temperature and vapour pressure at daily and monthly timescales (Jones et al., 2009). The AGCD addresses spatial and temporal gaps in the BoM station observations and provides useful estimates in remote areas where BoM observation networks are sparsely distributed. The AGCD of daily T_{max} and T_{min} with 5 km × 5 km resolution is available from 1900 onwards. Both ACCESS NWP forecasts and AGCD are gridded products from the BoM and utilise BoM observations in their generation. Aside from the variance in spatial resolution, the primary distinction between these gridded products lies in their derivation methods. AGCD is created by statistically assimilating historical observations, whereas ACCESS NWP is produced using numerical models that physically simulate the Earth's atmosphere and ocean conditions to forecast future weather. The AGCD can be requested directly from the BoM Climate Data Online (Bureau of Meteorology, 2022c) or NCI (NCI, 2022b).

3 | METHODS

In this section, we describe the details of each individual test and the method to combine them together.

3.1 | Domain test

Prior to undertaking any further tests, a domain test checks physical limits and physical conditions to identify obviously erroneous observations. In this study, daily T_{max} and T_{min} observations, denoted by T_{max} and T_{min}, are checked against the following condition

$$T_{min} < T_{max} < 60°C. \quad (1)$$

The lower limit of daily T_{min} is different for different types of stations:

- −30°C for mainland Australian stations;
- −40°C for mainland Australian stations above 1000 m.

The upper limit of daily T_{max} and the lower limit of daily T_{min} are taken from the current QC procedure implemented in the BoM. We will consistently update these values to align with any changes made by the BoM, particularly in the current changing climate. No further tests are carried out and a confidence score of zero is assigned if the condition in Equation (1) is not satisfied.

3.2 | Spatial regression test

The spatial regression test (Hubbard et al., 2005) is an effective spatial test to check the agreement of the

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observations from a target station of interest with the data from neighbouring high-quality stations. Traditionally, spatial tests are based on inverse distance methods that simply assign the weights to the values at surrounding stations according to the inverse of the distance separating the locations (Guttman & Quayle, 1990; Wade, 1987). Unlike inverse distance methods, the spatial regression test does not assign the nearest reference station with the greatest weight but assigns the weights based on the predictive skill of the neighbouring reference station relative to the target station. The spatial regression test implicitly accounts for the systematic difference caused by temperature lapse rate with elevation, which is an issue for the inverse distance method. Hubbard and You (2005) showed that the worst performance of the spatial regression test is still better than the best performance of the inverse distance method in different cases for daily $T_{\max}$ and that the largest improvements of the spatial regression test occur in the coastal and mountainous regions.

To perform the spatial regression test for a daily $T_{\max}$ (or $T_{\min}$) observation from a given target station located at s_0 and at time t_0, the nearest $N \leq 10$ reference stations located at $s_1, \ldots, s_N$ with a radius of 200 km from s_0 are selected. We denote the observed and true underlying daily $T_{\max}$ (or $T_{\min}$) at s_0 and at time t_0 by $y(s_0, t_0)$ and $y_T(s_0, t_0)$. The spatial regression test predicts the daily $T_{\max}$ (or $T_{\min}$) at s_0 and at time t_0, denoted by $y_{SRT}(s_0, t_0)$, by a linear combination of $y(s_j, t_0)$ $(j = 1, \ldots, N)$, which are the daily $T_{\max}$ (or $T_{\min}$) measured at $s_1, \ldots, s_N$ at time t_0:

$$y_{SRT}(s_0, t_0) = \beta_0 + \sum_{j=1}^{N} \beta_j y(s_j, t_0). \quad (2)$$

The parameters $\beta_0, \beta_1, \ldots, \beta_N$ are estimated by the least absolute shrinkage and selection operator (LASSO) method (Tibshirani, 1996), which minimises the following objective function

$$\sum_{t=1}^{T} \left\{ y(s_0, t) - \beta_0 - \sum_{j=1}^{N} \beta_j y(s_j, t) \right\}^2 + \lambda \sum_{j=0}^{N} |\beta_j|, \quad (3)$$

where λ is a prespecified penalty parameter (e.g., $\lambda = 0.01$ in this study) and $t = 1, \ldots, T$ is an independent period for parameter estimation. The use of LASSO instead of the standard least-squares estimation allows more robust parameter estimation in the presence of sporadic erroneous historical observations $y(s_0, t)$ for $t = 1, \ldots, T$. To simplify the notations, we do not distinguish the parameters and their estimates.

Given the estimate of the spatial regression method $y_{SRT}(s_0, t_0)$, $y_T(s_0, t_0)$ is assumed to follow a Gaussian distribution:

$$y_T(s_0, t_0) \mid y_{SRT}(s_0, t_0) \sim N\left(y_{SRT}(s_0, t_0) + \mu_{SRT}, \sigma^2_{SRT}\right), \quad (4)$$

where μ_{SRT} and σ_{SRT} are the bias and standard deviation of the estimate error and are estimated by robust statistical estimators based on the median and median absolute deviation (MAD):

$$\mu_{SRT} = \text{median}(y(s_0, t) - y_{SRT}(s_0, t)), \quad (5)$$

$$\sigma_{SRT} = \text{MAD}\left(y_{SRT}(s_0, t) - y(s_0, t) - \mu_{SRT}\right). \quad (6)$$

The MAD function in the equation above can calculated by

$$\text{MAD}(\epsilon_t) = k \times \text{median}\left(\left|\epsilon_t - \bar{\epsilon}\right|\right), \quad (7)$$

where $\epsilon_t = y_{SRT}(s_0, t) - y(s_0, t) - \mu_{SRT}$ is the model residuals, $\bar{\epsilon}$ is the average of ϵ_t and $k = 1.4826$ is a scale factor based on the Gaussian assumption. The parameter estimation does not require curated and error-free data (which are often unknown), because the robust parameter estimators given by Equations (5) and (6) as opposed to conventional and non-robust estimators, such as the mean and standard deviation of estimate error enables, can tolerate some errors and resist outliers in TPAWS observations (Rousseeuw & Croux, 1993). The model parameters of the spatial regression test, including the regression parameters $\beta_0, \beta_1, \ldots, \beta_N$ and the standard deviation parameter σ^2_{SRT}, are required to estimate for each target station against historical observations. The regression model in Equation (2) is assumed to be a global model and the regression parameters $\beta_0, \beta_1, \ldots, \beta_N$ are invariant to the time step t to increase sample sizes in estimating $N + 1$ parameters. With sufficient training data, we could assume that the regression parameters are dependent on calendar months to address the performance of the regression model being potentially different for different seasons/-months. Otherwise, with limited historical data for parameter estimation, we only assume the bias and standard deviation parameters μ_{SRT} and σ_{SRT} are monthly dependent. Therefore, there are 12 sets of μ_{SRT} and σ_{SRT} corresponding to 12 calendar months to address this potential seasonal dependence. Specially, only historical observations from the same calendar month as that of t_0 are used to estimate the model parameters corresponding to t_0 in Equations (5) and (6).

The confidence score is defined by a two-sided p-value in the framework of statistical hypothesis tests. The following hypothesis test is proposed to give a probability statement on how likely the observation $y(s_0, t_0)$

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is reliable:

$$H_0 : E(y_T(s_0, t_0)) = y(s_0, t_0),$$

$$H_1 : E(y_T(s_0, t_0)) \neq y(s_0, t_0).$$

The two-sided p-value p_2 can be expressed as a function of $y(s_0, t_0)$:

$$\begin{aligned} p_2 &= 2\min\{P(y_T(s_0, t_0) \leq y(s_0, t_0)), P(y_T(s_0, t_0) \geq y(s_0, t_0))\} \\ &= 1 - 2|p_1 - 0.5|, \end{aligned} \tag{8}$$

where $p_1 = P(y_T(s_0, t_0) \leq y(s_0, t_0))$ is the one-sided (left) p-value. Under the assumption given by Equation (4), we can derive the one-sided p-value for the spatial regression test as follows

$$p_1 = \Phi\left\{\sigma_{SRT}^{-1}\left(y(s_0, t_0) - \beta_0 - \sum_{j=1}^{N}\beta_j y(s_j, t_0) - \mu_{SRT}\right)\right\}. \tag{9}$$

The notation Φ in the above equation denotes the cumulative probability function of a standard Gaussian distribution. Finally, the confidence score of $y(s_0, t_0)$ based on the spatial regression test, CS_{SRT}, can be calculated by

$$\begin{aligned} CS_{SRT} = 1 - 2\Bigg|\Phi\Bigg\{\sigma_{SRT}^{-1}\Bigg(&y(s_0, t_0) - \beta_0 \\ &- \sum_{j=1}^{N}\beta_j y(s_j, t_0) - \mu_{SRT}\Bigg)\Bigg\} - 0.5\Bigg|. \end{aligned} \tag{10}$$

Conditional on the observations from the neighbouring stations, the confidence score represents the probability of the true underlying daily T_{max} (or T_{min}) being more extreme than the observed value from the target TPAWS. A greater confidence score indicates a more reliable observation. If $y(s_0, t_0)$ is close to $\hat{y}_{SRT}(s_0, t_0)$, which is the best guess of the spatial regression test, the confidence score will be close to 100%. If $y(s_0, t_0)$ is far from $y_{SRT}(s_0, t_0)$, the confidence score will be close to 0%. If $y(s_0, t_0)$ deviates from $y_{SRT}(s_0, t_0)$ by $3\sigma_{SRT}$, the confidence score will be equal to 0.27%.

The primary assumption of the spatial regression test is that the error of the spatial regression from the same calendar month follows a Gaussian distribution. We only undertake the spatial test if

1. At least two high-quality reference stations are located within 200 km from the target station;
2. A minimum of 300 complete historical observations in a 2-year period is required to estimate the model parameters;
3. The correlation between the observed and spatial predicted daily T_{max} (or T_{min}) is greater than 0.5 in the period used for parameter estimation.

3.3 | NWP test

The NWP test checks an observation of daily T_{max} (or T_{min}) from a TPAWS against the prediction from the ACCESS NWP product (described in Section 2). As all TPAWS considered in this study are located on the land, the corresponding prediction is taken from the nearest land-based ACCESS-G grid point. Because ACCESS-G only provides hourly T_{max} (or T_{min}) forecasts in the raw outputs, a pre-process is performed to derive daily T_{max} (or T_{min}) forecasts, which are used to directly compare with daily T_{max} (or T_{min}) observations from TPAWS. This pre-process attempts to use the most accurate hourly forecasts with the shortest lead time and consists of three steps: (1) the hourly T_{max} (or T_{min}) forecasts at the first six hours lead time for each forecast issued at 0000, 0600, 1200 and 1800 UTC to a single time series of hourly T_{max} (or T_{min}) forecast; (2) the UTC time in this time series is converted to the local time at a target location; (3) the daily T_{max} (or T_{min}) forecasts are taken from the maximum (or minimum) of the hourly T_{max} (or T_{min}) forecasts in the time series within 0900 AM–0900 AM at the local time.

Denote the derived daily T_{max} (or T_{min}) forecast of the ACCESS NWP product at a target station s_0 and time t_0 by $y_{NWP}(s_0, t_0)$. The primary assumption of the NWP test, similar to that made for the spatial regression test in Equation (4), is that $y_T(s_0, t_0)$ conditioning on $y_{NWP}(s_0, t_0)$ follows a Gaussian distribution:

$$y_T(s_0, t_0) \mid y_{NWP}(s_0, t_0) \sim N\left(y_{NWP}(s_0, t_0) + \mu_{NWP}, \sigma_{NWP}^2\right), \tag{11}$$

where μ_{NWP} is a bias correction parameter to address the bias of the NWP estimates and σ_{NWP} is a standard deviation parameter. To account for the possible seasonal dependence of the NWP estimate errors, the model parameters are assumed to be constant for each calendar month. Robust statistical estimation is adopted to minimise the impact of erroneous historical data as follows:

$$\mu_{NWP} = \text{median}(y(s_0, t) - y_{NWP}(s_0, t)), \tag{12}$$

$$\sigma_{NWP} = \text{MAD}(y(s_0, t) - y_{NWP}(s_0, t) - \mu_{NWP}), \tag{13}$$

where MAD is defined by Equation (7) and the median/-MAD function is calculated over all t in the estimation period with the same calendar month as t_0. Using the same argument for the spatial regression test and the distribution assumption given in Equation (11), we define the confidence score of the NWP test, CS_{NWP}, as

$$\begin{aligned} CS_{NWP} = 1 - 2 \\ \times \left|\Phi\left\{\sigma_{NWP}^{-1}(y(s_0, t) - y_{NWP}(s_0, t) - \mu_{NWP})\right\} - 0.5\right|. \end{aligned} \tag{14}$$

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In addition to the distribution assumption in Equation (11), the following two assumptions are required for the NWP test:

1. A minimum of 300 complete historical observations in a two-year period is required to estimate the model parameters;
2. The correlation between the observed and NWP estimated daily $T_{\max}$ (or $T_{\min}$) is greater than 0.5 in the period used for parameter estimation.

3.4 | AGCD test

The AGCD test checks an observation of daily $T_{\max}$ (or $T_{\min}$) from a TPAWS against the estimate from the AGCD (described in Section 2). Unlike the ACCESS NWP product, the AGCD provides the estimate of daily (instead of hourly) $T_{\max}$ (or $T_{\min}$) and no pre-processing is needed. The AGCD test essentially is the same as the NWP test, and the only difference is that the NWP estimate y_{NWP} is replaced by the AGCD estimate y_{AGCD}. We make the distribution assumption as in Equation (11), estimate the bias correction and standard deviation parameters for each month by Equations (12) and (13) and calculate the confidence score by Equation (14). The assumptions and minimum requirements of the NWP test are applied to the AGCD test too. Therefore, we do not repeat all the details in the paper.

3.5 | Merged test

Each individual test (e.g., spatial regression, NWP or AGCD test) provides a probabilistic prediction of daily $T_{\max}$ (or $T_{\min}$) and issues a confidence score of an observation from a TPAWS based on different sources of information. A merged test combines the prediction from individual tests and provides a confidence level as an overall assessment. Because the predictive distribution of each test is a Gaussian distribution, we use a Gaussian mixture model to combine them together for a unified distribution. This approach can be considered as a simplified version of Bayesian model averaging (Hoeting et al., 1999) without specifying an informative prior. With given information F_i, the ith test assumes that the distribution of $y_{\text{T}}(s_0, t_0)$ conditioning on F_i follows a Gaussian distribution

$$y_{\text{T}}(s_0, t_0) \mid F_i \sim N\left(\mu_i, \sigma_i^2\right). \tag{15}$$

The above equation is a generic form of Equations (4) and (11) specified for the spatial regression test and the NWP test. The merged test assumes that the distribution of $y_{\text{T}}(s_0, t_0)$ conditioning on $F_1, F_2, \ldots, F_n$ follows a Gaussian mixture model

$$y_{\text{T}}(s_0, t_0)|F_1, F_2, .., F_n \sim \text{MN}\left(\mu_1, \ldots, \mu_n, \sigma_1^2, \ldots, \sigma_n^2, w_1, \ldots, w_n\right), \tag{16}$$

where $\text{MN}\left(\mu_1, \ldots, \mu_n, \sigma_1^2, \ldots, \sigma_n^2, w_1, \ldots, w_n\right)$ represents a mixture of n Gaussian distributions $N\left(\mu_1, \sigma_1^2\right), \ldots, N\left(\mu_n, \sigma_n^2\right)$ with weights $w_1, \ldots, w_n$. In this study, $n = 3$ corresponding to three individual tests. As the mean and standard deviation parameters of the Gaussian mixture model have been estimated by each individual method, the weight parameters are the only unknown parameters in the model. Furthermore, the weight parameters are assumed to be monthly dependent, reflecting that the performance of each method may be different from month to month.

The weight parameters are estimated by a robustification of the expectation–maximisation (EM) algorithm (Deveaux & Krieger, 1990). This robust estimation seeks to find the maximum likelihood estimate by iteratively applying the following two steps:

- *Expectation step* (*E step*): compute $Q\left(w|w^{(p)}\right)$, the expectation of the log-likelihood function of $w = (w_1, \ldots, w_{n-1}, 1 - w_1 - \ldots - w_{n-1})$ given the current estimate $w^{(p)} = \left(w_1^{(p)}, \ldots, w_n^{(p)}\right)$, by

$$Q\left(w|w^{(p)}\right) = \sum_{t=1}^{T}\sum_{i=1}^{n} z_t(i)\left[\log(w_i) + \log\left\{f_i(y(s_0, t))\right\}\right], \tag{17}$$

where

$$z_t(i) = \frac{w_i^{(p)} f_i(y(s_0, t))}{\sum_{j=1}^{n} w_j^{(p)} f_j(y(s_0, t)),}, \tag{18}$$

$$\log\left\{f_i(y(s_0, t))\right\} = -2\log(\sigma_i) - \sigma_i^{-1}|y(s_0, t) - \mu_i|. \tag{19}$$

- *Maximization step* (*M-step*): find the parameters $w^{(p+1)}$ that maximize the quantity $Q\left(w|w^{(p)}\right)$.

  In the above expectation/maximization algorithm, the restriction $\sum_{i=1}^{n} w_i = 1$ is required to ensure a proper merged test.

We use the same argument made for the spatial regression test (Section 3.2) and define the confidence score in the framework of hypothesis tests. Based on the cumulative distribution function of the Gaussian mixture distribution, the one-sided left-tail p-value p_1 for the merged test can be expressed as

$$p_1 = \sum_{i=1}^{n} \text{w}_\text{i}\Phi\left\{\sigma_i^{-1}(y(s_0, t_0) - \mu_i)\right\}. \tag{20}$$

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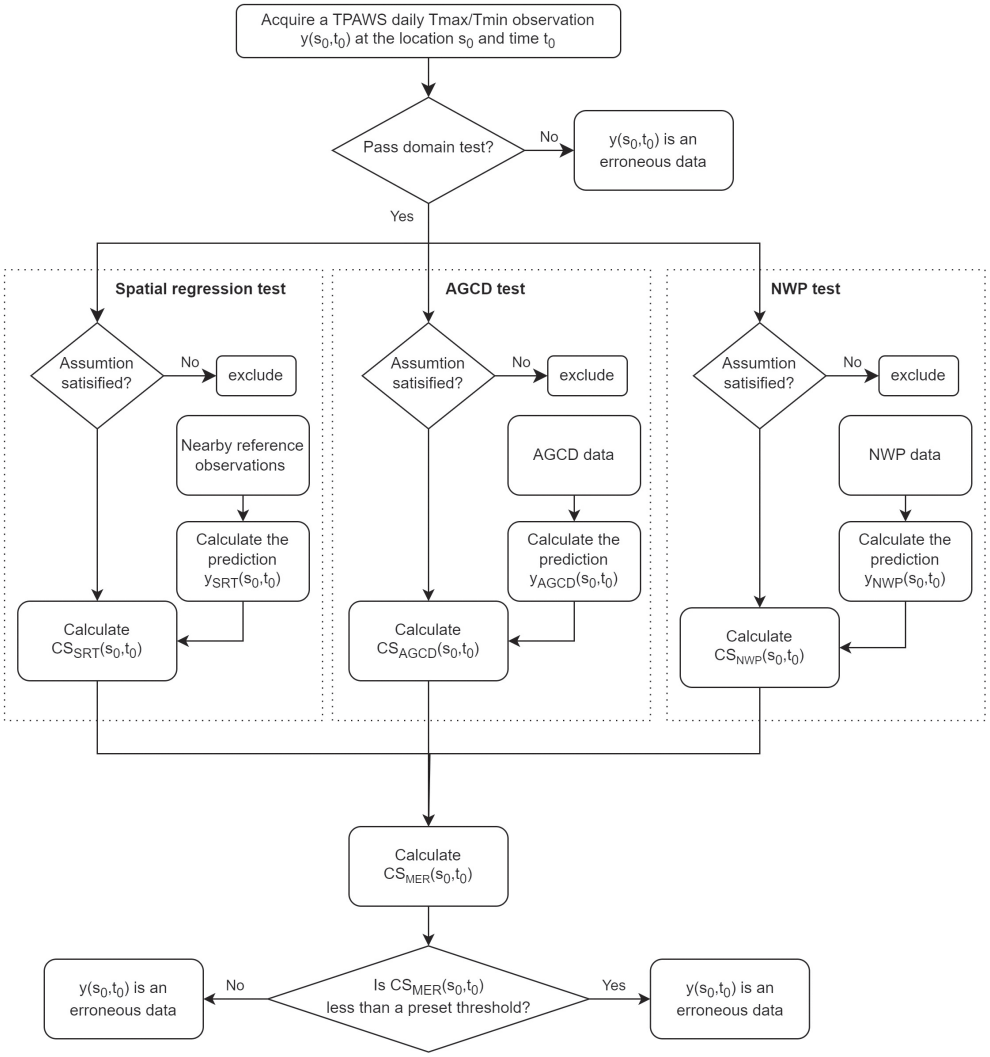


FIGURE 1 A flow diagram detailing the utilization of various data sources as inputs for distinct test method and the subsequent integration into the merged test.

The prediction from the merged test methos μ_{MER} is taken from the weighted average of the predictions from individual tests:

$$\mu_{\mathrm{MER}} = \sum_{i=1}^{n} \mathrm{w_i}\mu_{i.}. \tag{21}$$

Therefore, we use the relationship between the one-sided left-tail p-value and the two-sided p-value in Equation (6) and obtain the confidence score for the merged test $\mathrm{CS_{MER}}$ as

$$\mathrm{CS_{MER}} = 1 - 2\left|\sum_{i=1}^{n} \mathrm{w_i}\Phi\left\{\sigma_i^{-1}(y(s_0,t_0) - \mu_i)\right\} - 0.5\right|. \tag{22}$$

Although the one-sided p-value for the merged test is a weighted average of that for each individual test, the confidence score defined by the two-sided p-value is not a weight average of that for each test. Figure 1 illustrates a flow diagram detailing the utilisation of various data sources as inputs for distinct tests and the subsequent integration of these tests into a merged assessment for final analysis.

4 | VALIDATION WITH A SYNTHETIC DATASET BASED ON HIGH-QUALITY STATIONS

The performance of the proposed test method is assessed against two synthetic datasets (one for $T_{\max}$ and the other for $T_{\min}$) in lieu of real TPAWS observations, whose quality is unknown. The synthetic datasets are based on the daily $T_{\max}$ and $T_{\min}$ observations between 1 January 2017 and 31 December 2018 from 100 high-quality stations from the Australian Climate Observations Reference Network (ACORN) Surface Air Temperature (SAT) dataset (Bureau of Meteorology, 2022b). The selection of the test stations for daily $T_{\max}$ and $T_{\min}$ is mostly the same, while seven of them are different to include more observations of

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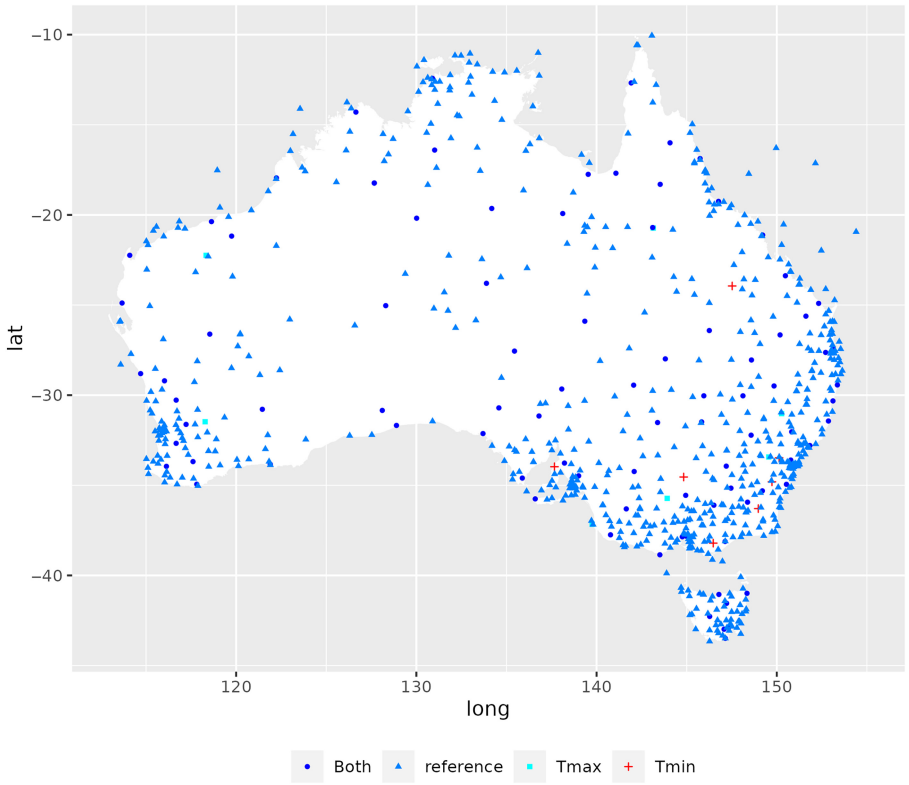


FIGURE 2 The spatial distributions of test stations for both T_{max} and T_{min}, T_{max} only and T_{min} only and reference stations used for the spatial regression test.

extremely hot and cold weather conditions for daily T_{max} and T_{min}, respectively. The full list of the test stations is provided in Table S1 and the spatial distribution of these test stations and the reference BoM stations used for the spatial regression test is illustrated in Figure 2.

The synthetic dataset is generated by inserting random errors to 1953 (i.e., ~2.7%) daily T_{max} observations out of totally 72,254 observations (i.e., two-year daily data for 100 stations excluding a few missing values). For daily T_{min}, 1956 (i.e., ~2.7%) out of totally 72,472 observations are inserted with random errors. The inserted random error for both daily T_{max} and T_{min} is uniformly distributed between [−6°C, −2°C] or [2°C, 6°C] across all stations. Negative (or positive) inserted errors simulate the situation when TPAWS observations are greater (or less) than ground truth. The structure of random perturbation is chosen by our understanding of potential users' expectation. We refer to observations with inserted errors as modified observations and the rest as unmodified observations. None of test data fail the domain test and we focus on our discussion on the spatial regression, NWP, AGCD tests and the merged test. It is potentially challenging for the test method to perform well at some 'difficult' test stations, including 16 remote stations with less than three reference stations in a radius of 200 km and 27 coastal stations within 10 km from the costal line for both the daily T_{max} and T_{min} synthetic datasets.

The performance of our proposed test is evaluated by the leave-one-out cross-validation scheme as following. For each target observation on a given day, the test parameters are calibrated by using all the observations on the rest of days and corresponding observations from nearest stations with a radius of 200 km, NWP data and AGCD data. The calibrated parameters are then used to derive the confidence score for the target observation. This procedure is repeated for every observation in the entire dataset (e.g., 72,254 observations for T_{max} and 72,472 observations for T_{min}) and their confidence scores are summarised to obtain the overall performance assessment.

Table 1 shows the average parameter estimates of each test method pooled from all test stations and dates for the synthetic dataset. (Figure S2 further shows the distributions of parameter estimates.) The AGCD test, with the smallest standard deviation estimate, is proven to be the most accurate estimate of daily T_{max} and T_{min} and is therefore assigned the largest weights in most cases when combining methods together. In contrast, the weights for the NWP test are often negligible because weather forecasts provide far less accurate information than neighbouring reference stations. Comparing T_{max} with T_{min}, we found that the standard deviation parameters of any test for T_{max} are consistently greater than the corresponding ones for T_{min}. This suggests that it is more difficult to predict daily T_{min} than daily T_{max}.

The proposed method produces the confidence score rather than a dichotomous classification of the quality of the observation. Classification can be performed by defining a threshold on the confidence score. This threshold

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TABLE 1 The average parameter estimates across all stations and dates for different test methods.

		T_{max}			T_{min}		
Data	**Parameters**	**Spatial**	**NWP**	**AGCD**	**Spatial**	**NWP**	**AGCD**
Synthetic data	μ	0.01	−1.15	0.03	0.03	0.37	0.07
	σ	0.70	1.01	0.33	1.04	1.47	0.54
	w	0.16	0.02	0.82	0.21	0.02	0.77
DPIRD data	μ	0.02	−1.31	−0.23	−0.02	0.23	0.32
	σ	0.61	0.95	0.72	0.92	1.11	1.17
	w	0.60	0.04	0.36	0.69	0.15	0.16

Abbreviations: AGCD, Australian Gridded Climate Data; DPIRD, department of Primary Industries and Regional Development; NWP, numerical weather prediction.

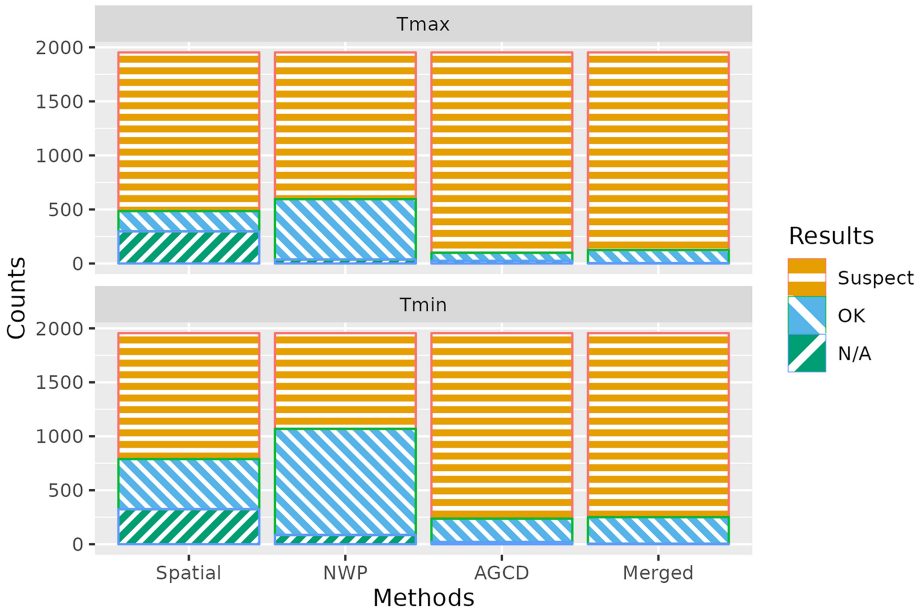


FIGURE 3 The number of observations identified as suspect (i.e., the confidence score is less than a threshold), OK (i.e., the confidence score is greater than a threshold) or not applicable (N/A) (i.e., the test method is not appliable for a target observation) when the confidence score threshold is 0.27%.

value is left for the users to select based on their particular application and risk preference. To validate the test method quantitively, we choose to classify of confidence scores into error flags and evaluate the predicted classification. The common approach may be to use the 1, 2 or 3 standard deviation as a threshold, which corresponds to a 31.7%, 4.5% or 0.27% threshold, respectively. Figure 3 shows the number of suspect observations identified by each test method if the confidence score threshold is 0.27%. In contrast to each individual test that can be limited to some but not all observations due to data availability and model assumptions, the merged test can complement to each single test and be applicable for all observations in this synthetic dataset. Hit rates (i.e., the number of the modified observations with confidence scores less than the threshold divided by the total number of the modified observations) and false alarm rates (i.e., the number of the unmodified observations with confidence scores less than the threshold divided by the total number of the unmodified observations) are used to measure the performance of test methods. Figure 4 shows the sensitivity of hit rates and false alarm rates to the threshold. As expected, hit rates decrease and false alarm rates increase as the threshold increases. The threshold based on three standard deviations leads to a balanced performance between hit rates and false alarm rates and therefore we fix the threshold as 0.27% for the rest of this study. With the same threshold, the hit rates for T_{max} are greater than those for T_{min}, and the alarm rates are very similar for two variables.

The overall performance of spatial, NWP, AGCD and merged tests for daily T_{max} is provided in Table 2. The AGCD test achieves the highest hit rates (followed by the spatial and NWP tests) but the highest false alarm rates as well. The merged test, combining three individual tests, effectively reduces the false alarm rate calculated for all observations while maintaining a hit rate similar to that of

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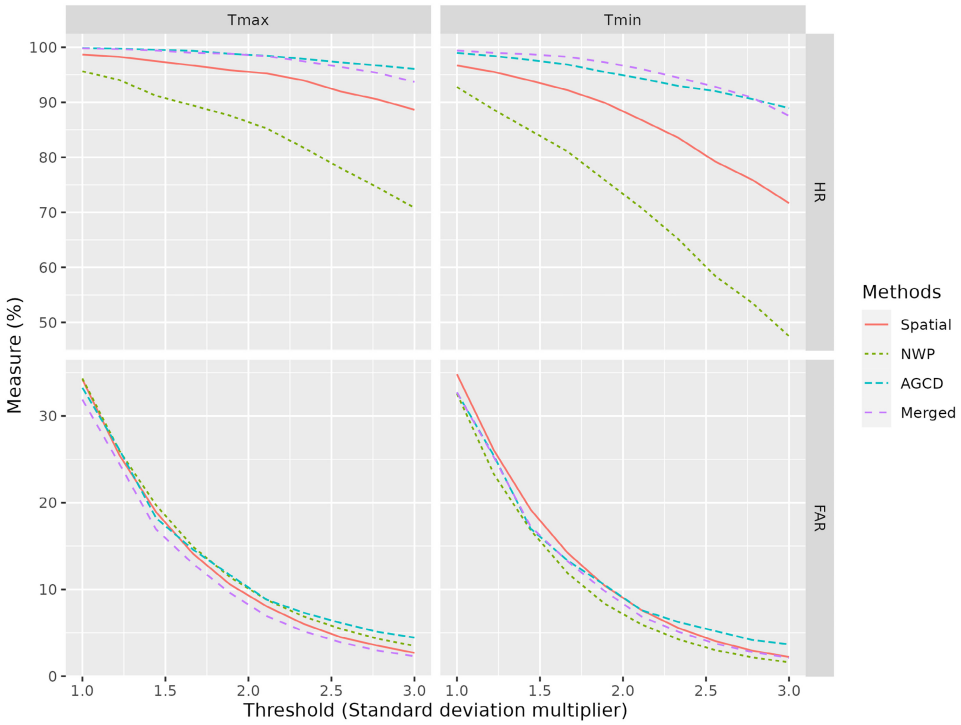


FIGURE 4 Hit rates (HR) and false alarm rates (FAR) as a function of confidence score thresholds for different test methods.

TABLE 2 Summary statistics for the test performance of the spatial regression, NWP, AGCD and merged tests applied to daily T_{max} when the confidence score threshold is chosen to be 0.27%.

Measure	Category	Spatial	NWP	AGCD	Merged
Hit rate (%)	All observations	88.6	70.8	96.1	93.7
	<20°C	95.0	75.7	95.5	96.4
	20–35°C	87.9	71.0	95.7	93.3
	>35°C	81.0	63.3	98.1	91.5
False alarm rate (%)	All observations	2.7	3.5	4.4	2.3
	<20°C	2.5	2.3	3.5	2.2
	20–35°C	2.8	3.9	4.6	2.4
	>35°C	2.4	3.6	5.4	2.2
The percentage of stations with >80% hit rates and <10% false alarm rates (%)		80.9	28	86	91

Abbreviations: AGCD, Australian Gridded Climate Data; NWP, numerical weather prediction.

the AGCD test. Table 2 also presents the test performance for cold (<20°C), normal (20–35°C) and hot (>35°C) days. As the distribution of daily T_{max} is skewed to the right, it is more difficult for any method to perform well for hotter days. Nevertheless, the merged test can achieve more than 90% hit rates and less tan3% false alarm rates for all categories. The impact of inserted errors on the hit rates is investigated in Figure 5. As expected, the greater the magnitude of inserted errors is, the greater hit rate any method can obtain. The merged test can detect an inserted error (positive or negative) with a magnitude higher than 3°C in more than 80% of cases. Modified observations with negative inserted errors are more easily detected by all tests. The overall test performance for daily T_{min} is given in Table 3 and the hit rates are significantly smaller for daily T_{min} than for daily T_{max}. The hit rates for daily T_{min} can be even smaller in extremely cold days than other days.

Figures 6 and 7 present the station-wise hit rates and false alarm rates. The spatial regression test can only be applied to 84 stations because there are insufficient neighbouring stations for the remaining 16 stations. The other two individual tests and the merged test can be used for all test stations. The stations with fewer than 80% hit rates for the merged test are either close to the costal line or located in North Queensland, which is hot and humid with a distinct climate. The false alarm rate for the merged test

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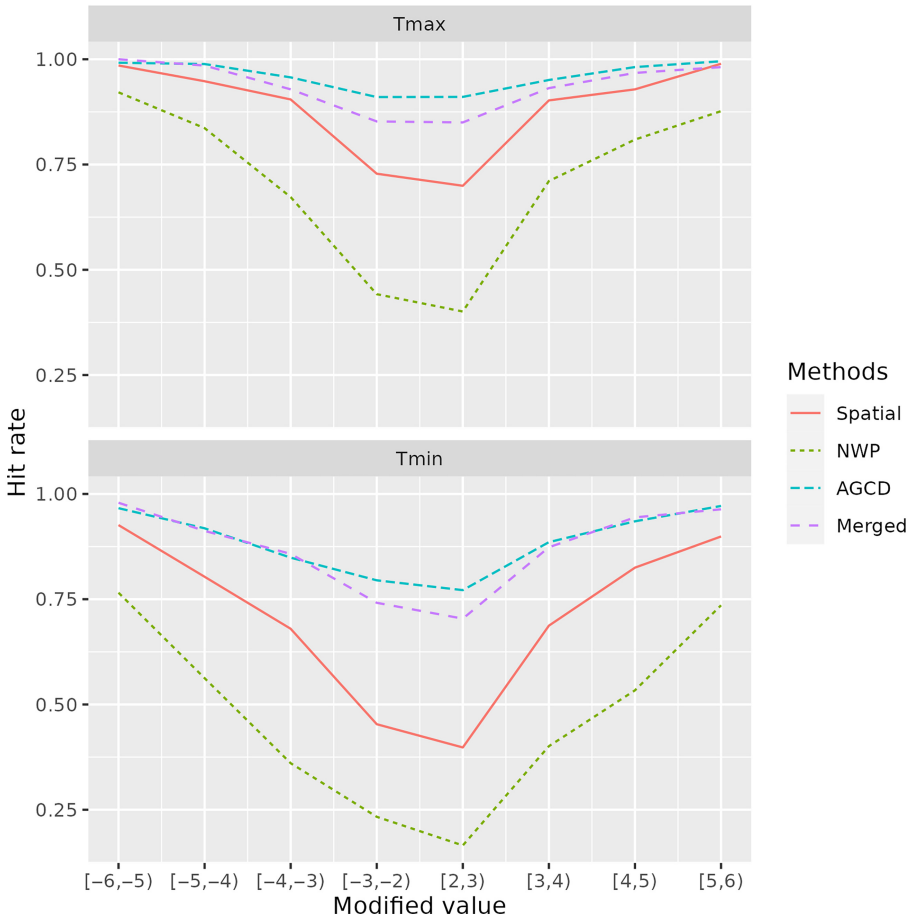


FIGURE 5 Hit rates as a function of the inserted errors for different test methods.

TABLE 3 Summary statistics for the test performance of the spatial regression, NWP, AGCD and merged tests applied to daily T_{min} when the confidence score threshold is chosen to be 0.27%.

Measure	Category	Spatial	NWP	AGCD	Merged
Hit rate (%)	All observations	71.6	47.4	88.8	87.4
	<0°C	63.5	36.8	75.0	80.9
	0–10°C	67.0	39.1	82.6	84.2
	>10°C	74.5	52.4	92.6	89.3
False alarm rate (%)	All observations	2.2	1.6	3.7	2.1
	<0°C	1.6	0.6	2.5	1.5
	0–10°C	2.1	1.2	3.4	2.0
	>10°C	2.3	1.9	3.9	2.2
The percentage of stations with >80% hit rates and <10% false alarm rates (%)		36.9	7	71	73

Abbreviations: AGCD, Australian Gridded Climate Data; NWP, numerical weather prediction.

is less than 10% for any test station. The false alarm rate for the AGCD test can be rather high for central Australia, where the BoM observation network is sparse. For daily T_{min}, the hit rates of the merged test are small in Tasmania and southeast New South Wales, where extremely cold days occur more often.

5 | A CASE STUDY BASED ON DPIRD DATA

We apply the test method to evaluate the quality of daily T_{max} and T_{min} observations from 184 DPIRD TPAWS which are all located in WA. As DPIRD TPAWS are

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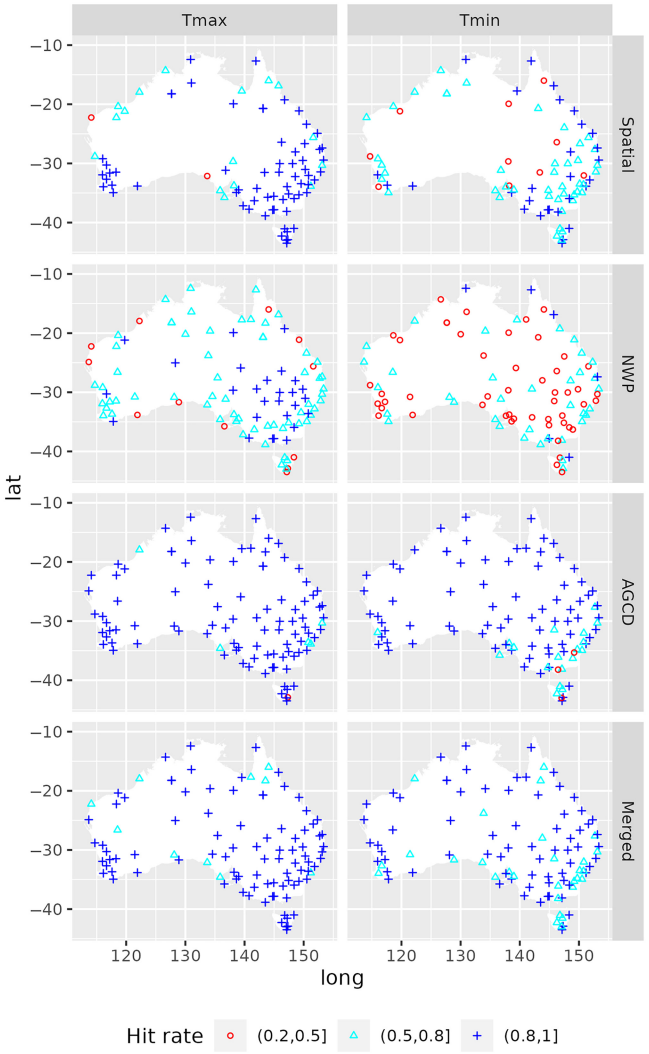


FIGURE 6 Station-wise hit rates for different test methods.

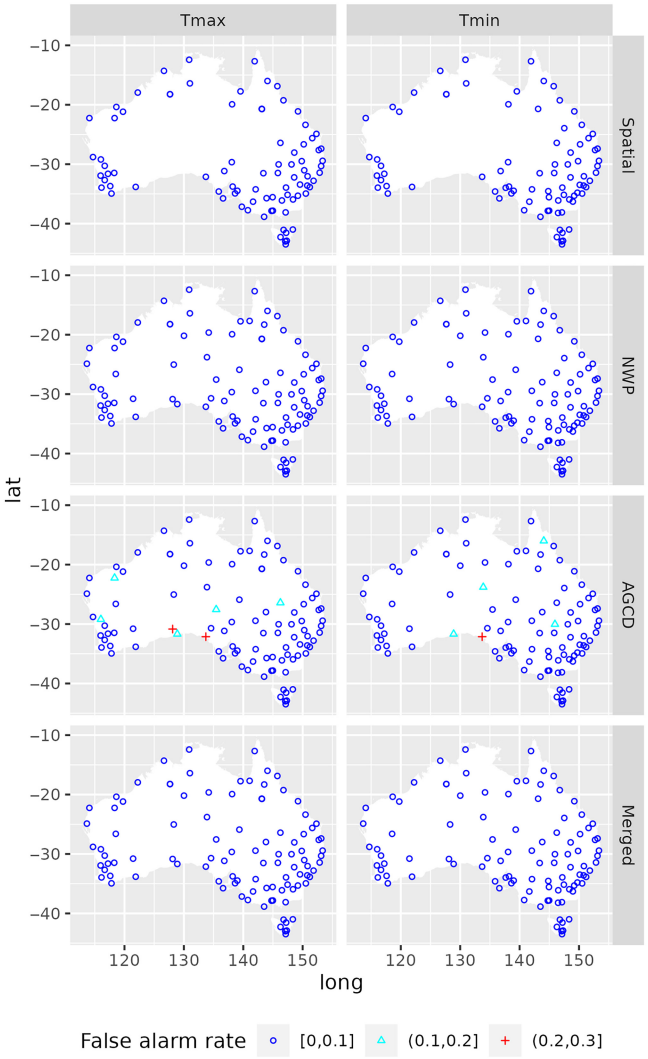


FIGURE 7 Station-wise false alarm rates for different test methods.

managed by a state government agency, we presume that they are equipped with better sensors than personal TPAWS and was installed and served by professionals. As a result, we do not expect a high percentage of errors in the data. The evaluation period is from 1 January 2019 to 31 December 2019 and the model parameters are estimated in a separate period from 1 January 2017 to 31 December 2018. In total, after excluding missing values, we evaluate 62,440 DPIRD daily T_{max} and 68,759 T_{min} observations. The metadata of the DPIRD stations are provided in Table S2. The DPIRD weather data can be accessed via an Application Programming Interface (API) (DPIRD, 2022b). All test data satisfy the condition specified in the domain test and the spatial regression test can be applied to all test stations with sufficient neighbouring BoM reference stations.

The average parameter estimates of each test method for the DPIRD dataset are provided in Table 1. (Figure S3 illustrates the distribution of parameter estimates for the DPIRD dataset). The predictions of the spatial regression test and the AGCD test are unbiased overall, but the NWP test underestimates daily T_{max} by 1.3°C on average. For the DPIRD data, the spatial regression test leads to the least overall prediction uncertainty (indicated by the smallest value of the average standard deviation parameter estimate) and therefore is assigned the largest weight.

The merged test detects 1431 (2.3%) for T_{max} and 2698 (4.4%) for T_{min} potentially erroneous observations with the confidence score less than 0.27% in the entire test dataset. The station-wise proportion of potential erroneous observations varies from 0% to 6.3% for T_{max} and from 0.8% to 12.3% for T_{min} (Figure 8). No obvious spatial pattern is shown in Figure 8, except that slightly more possibly erroneous data are identified at those stations on the south coast of WA. Additionally, we observed 542 instances for T_{max} and 2049 for T_{min}, where the data deviated from the

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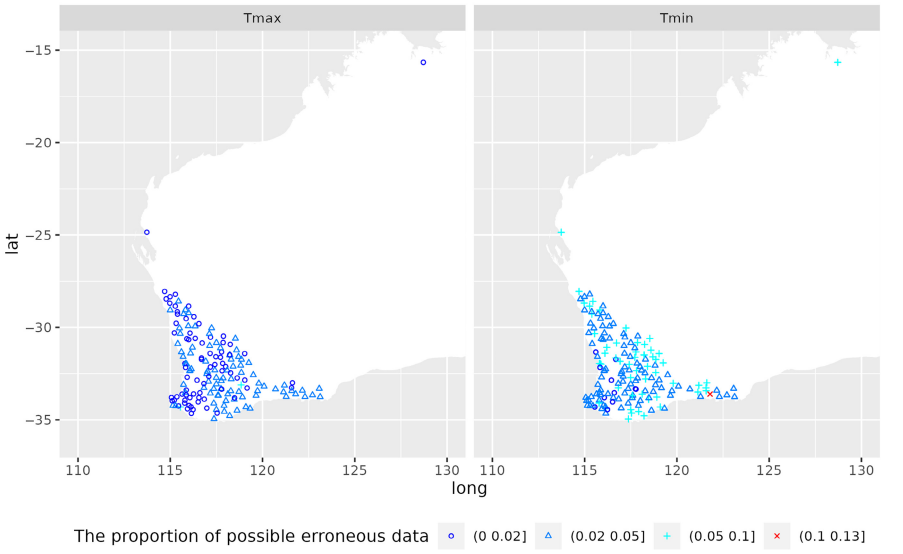


FIGURE 8 The spatial distribution of the proportion of possible erroneous data at each individual Department of Primary Industries and Regional Development station.

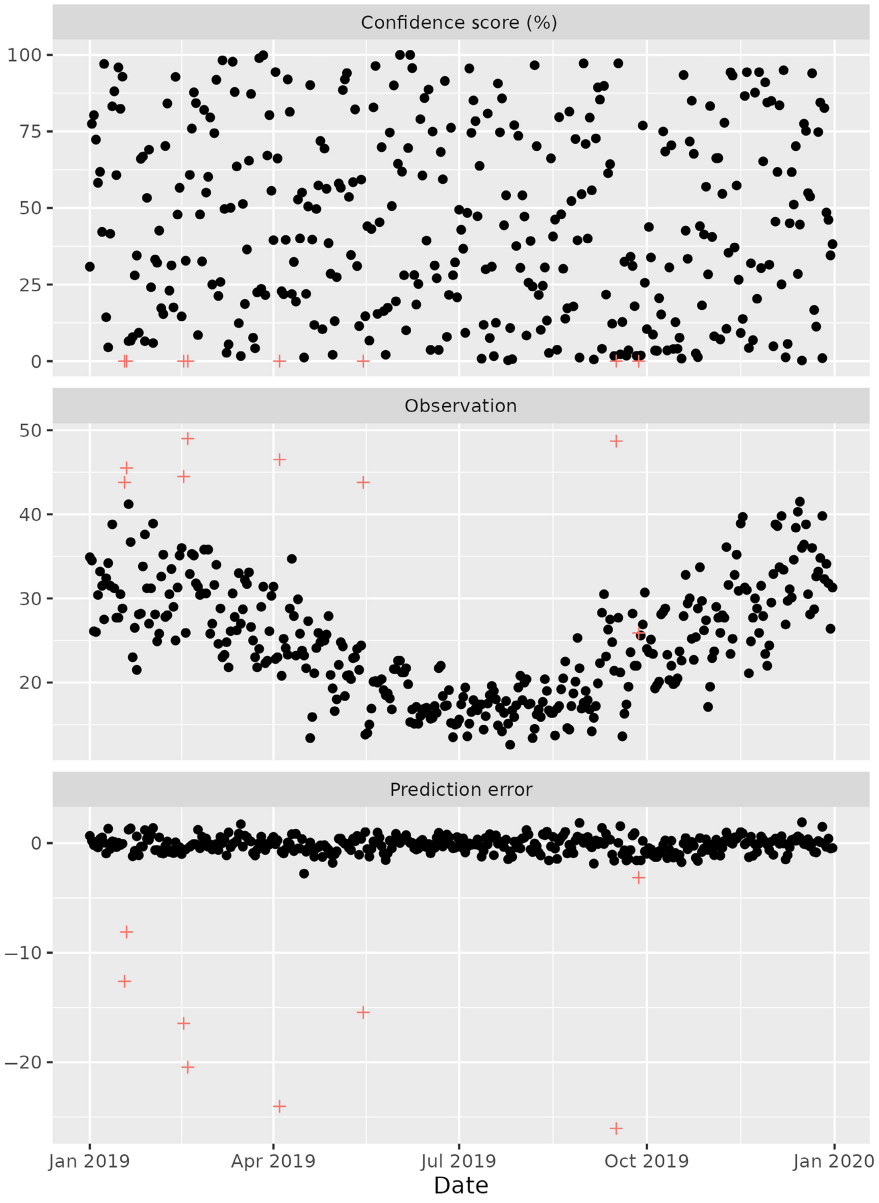


FIGURE 9 The confidence score, daily T_{max} observations and the prediction error of the merged test for station NY002.

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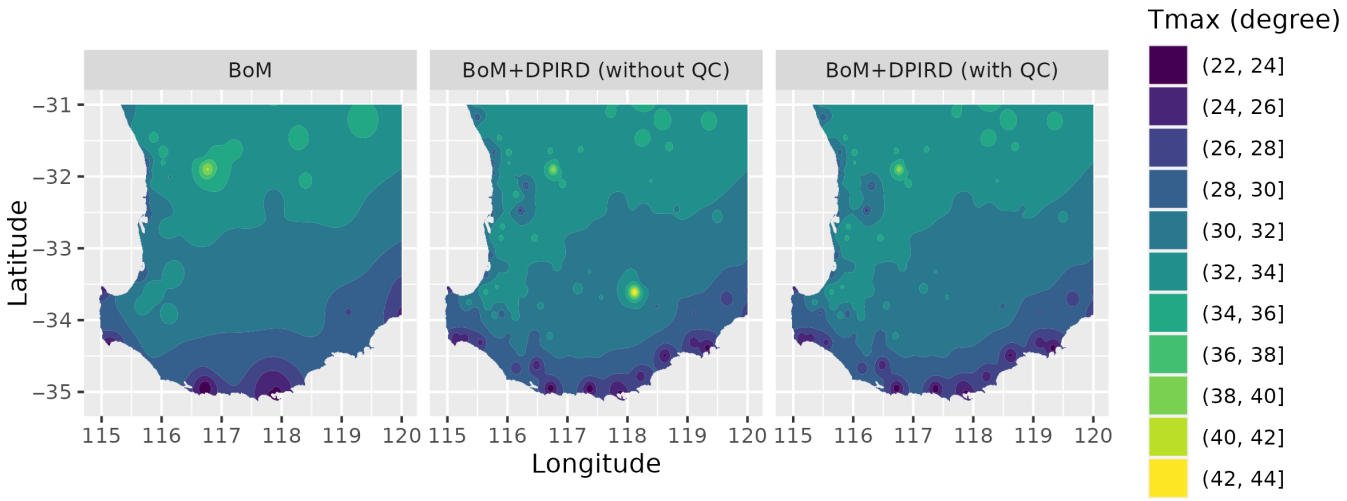


FIGURE 10 High resolution (1 km × 1 km) T_{max} estimates using the inverse distance weighting for 18 January 2019 based on different datasets (Bureau of Meteorology [BoM], BoM and Department of Primary Industries and Regional Development [DPIRD] without quality control, BoM and DPIRD with quality control [QC] in panels from left to right).

predictions of the merged test by more than 2°C. Given that all DPIRD stations are situated in agricultural regions, this 2°C difference might signal a higher level of suspicion regarding these discrepancies. It is important to note that the 2°C rule of thumb might not be as applicable for TPAWS in urban areas, where the urban heat island effect could potentially influence the observations by more than 2°C.

Figure 9 illustrates the assessment based on the merged test of daily T_{max} for a specific station, NY002, situated in the southwest region of WA. The confidence scores span approximately uniformly across [0, 1], encompassing eight observations within a year of daily data (highlighted in red) that are flagged as potentially erroneous. Detailed evaluations for these eight observations are outlined in Table S3. Some potentially erroneous data points are discernible merely by examining the time series trend. For instance, the daily T_{max} reading of 43.8°C on 15 May 2019 is highly improbable during autumn in the southwest region. However, certain other potentially erroneous data are not as evident when observed solely through the lens of the time series. Take, for instance, the daily T_{max} observation of 25.9°C on 27 September 2019, which might seem reasonable when compared to the observations on adjacent days at the same station. Nevertheless, all neighbouring stations report significantly lower daily T_{max} values on that day (refer to Figure S4), prompting its identification via the spatial regression test. This highlights the strength of our approach in evaluating the quality of daily T_{max}/T_{min} from TPAWS.

TPAWS potentially enhances the accuracy of daily T_{max}/T_{min} estimation on a high-resolution grid by augmenting the current weather observation network's density. Figure 10 illustrates T_{max} estimates for 18 January 2019 in southwest Western Australia at 1 km × 1 km grid points using a straightforward inverse distance weighted interpolation method with different datasets. The figure presents three datasets (BoM only, BoM and DPIRD without QC, BoM and DPIRD with QC) in panels from left to right. It is evident that the estimates derived from BoM and DPIRD datasets can capture more localised features compared to those relying solely on BoM data. Moreover, employing QC on BoM and DPIRD data eliminates the influence of an erroneous observation of 43.8 degrees at the NY002 station. Our ongoing investigation delves into the value of integrating TPAWS observations to enhance the accuracy of T_{max}/T_{min} estimates. In future research, we will conduct sensitivity analyses on confidence score thresholds and interpolation methods to further refine these estimations.

6 | CONCLUSIONS AND DISCUSSION

TPAWS provide a compelling data source for scientists and practitioners to obtain and estimate more accurate fine-scale atmospheric conditions (including daily T_{max} and T_{min}) than the current primary weather observation network can offer. However, the observations from TPAWS may come with some level of uncertainty inadequate owing to a lack of quality control. In this study, we develop a statistical approach to evaluate the quality of daily T_{max} and T_{min} observations collected from TPAWS in Australia by using the trusted data source from the BoM. Our framework consists of constructing individual tests

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for each BoM data source and merging all these individual tests to provide a final assessment on TPAWS observations. Each individual test issues a prediction of daily T_{max} and T_{min} at a target TPAWS and quantifies the associated prediction uncertainty. The merged test combines the test results based on the Gaussian mixture model and can be applied to the situation where a certain type of reference data are missing. The overall assessment is made in the form of p-value-based confidence scores that measure the difference between the target observation and trusted reference data. Our method is validated by synthetic datasets based on BoM high-quality observations and is applied to daily T_{max} and T_{min} observations from 184 AWS owned by the DPIRD of Western Australia. Based on the findings from the case studies, we draw the following conclusions:

1. The merged test can effectively detect modified data with inserted errors of more than 2°C magnitude.
2. The merged test leads to smaller false alarm rates on average than any individual test and hit rates close to the greatest hit rates from individual tests.
3. The merged test tends to perform worse in regional areas where primary observation networks are distributed sparsely.
4. The proposed confidence score is a good measure of the quality of daily T_{max} and T_{min} observations.
5. It is more difficult to detect and confirm erroneous observations of daily T_{min} than those of daily T_{max}.

The model framework is automatic, simple and expandable to handle a number of different available trusted data sources and interpretable in the sense that users can trace back where the disagreement is between TPAWS observations and the trusted data sources. The framework can be readily applied to other regions with different trusted data sources. Due to the AGCD dataset's exclusive availability from the BoM observations in Australia, replicating the results outlined in Section 4 with different data sources in other countries may not be straightforward. Furthermore, the real-world scenario depicted in Section 5, where AGCD carries comparable weight to NWP, is confined to Australia, representing a limitation of our study. Although the framework can be generalised to include different official data, it requires dedicated efforts to select suitable, reliable reference datasets and assess the model performance across various countries worldwide.

Our approach is tailored to detect erroneous observations (specifically from AWS) instead of to filter out legitimate extreme temperature values. The proposed method not only identifies potential errors but also estimates the magnitude of these errors through the predictions generated from merged tests. As our methodology operates on statistical principles, attributing specific causes to high or low temperature errors is not within the scope of current framework.

The most fundamental assumption of the proposed method is that erroneous data are present in the form of outliers in TPAWS observations. The test method may not work effectively as demonstrated in this study if this assumption is invalid. For example, the test method cannot detect systematic biases. We would recommend implementing appropriate QA procedures, particularly for private TPAWS, prior to carrying out the proposed QC method.

Synthetic data with a more complex error structure will be considered in future work. For example, we could further assume that inserted random perturbations are temporally correlated to simulate the situation where a sensor failure results in erroneous measurements on multiple days in a row. Nevertheless, we would expect similar test performance even with temporally correlated errors, as we indeed identified two instances where erroneous data happen in two consecutive days in our real-data application (e.g., Figure 9).

The spatial regression test could be extended to a spatial–temporal test. In the early development of our test method, we trailed a few spatial–temporal tests during the development of the framework (Shao et al., 2021a, 2021b) and found that none of them overperformed the current spatial regression test. This may be caused by the fact that more complex models are associated with more model parameters and require more historical observations for reliable parameter estimation. It may be more worthwhile to consider spatial–temporal tests for subdaily climate observations that have a stronger temporal autocorrelation and more abundant training data. In addition, we would focus on developing parsimonious test models that greatly ease the burden of algorithm implementation, deployment and maintenance in an upcoming operational QC system.

We utilise NWP data due to their ability to offer valuable insights into local weather conditions, especially in remote regions, leveraging the perspective of numerical climate models. Acknowledging that NWP data generally come with relatively low spatial resolution, our algorithm tends to automatically assign larger standard deviation parameters and smaller weights when merging NWP tests than in spatial regression tests (using BoM observations) and AGCD tests (using AGCD data). While we do not explicitly specify the minimum spatial resolution for gridded data, any gridded data with a coarse resolution will inherently receive smaller weights in the merged test.

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The proposed test framework, initially designed for daily T_{max} and T_{min}, may be applicable to other climate variables, but it is not a straightforward task. For example, our initial investigation found that the test method in the current format may not be appropriate for daily rainfall, which is the most important climate variable for many applications. Further model development is required to overcome the mixture of discrete and continuous distributions (due to many zero rainfall) and heterogeneous errors from reference datasets.

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DATA AVAILABILITY STATEMENT

The BoM weather observations can be requested from the BoM Climate Data Online (http://www.bom.gov.au/climate/data/). The ACCESS NWP APS2 forecast data can be directly downloaded from National Computational Infrastructure (NCI) with an OpenDAP server option (https://doi.org/10.25914/608a99ac28fa1). The AGCD can be requested directly from NCI (https://doi.org/10.25914/6009600304b02). The DPIRD weather data can be accessed via an Application Programming Interface (API) (https://weather.agric.wa.gov.au/developer-api).

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